Supplement to: Testing a predictive biomarker under drug carryover in aggregated N-of-1 trials: a staged simulation study

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This supplement gives the tables that support the main text but are not needed to follow it. The analyses are labeled A0 to A4 as in the definition table of the main text: A0 is the starting analysis, A1 the binary indicator with the working-covariance standard error, A2 the binary indicator with the cluster-robust standard error (CR2), A3 the exposure-weighted predictor at the true half-life with CR2 (an oracle), and A4 the exposure-weighted predictor at the half-life chosen by AIC on the model without the interaction, with CR2. The companion papers call A4 G10. The scenario families are those of the main text. Every table is computed from stored simulation output by the same code as the main text (`shared-data.R`).

Rung 0 to rung 1: the starting analysis

Table S1 gives Type I error at the six null cells of the base grid. Table S2 separates the three differences between A0 and A1.

Table S1: Type I error of the five analyses at the six null cells of the corrected grid ($c_{bm} = 0$, $t_{1/2} = 1$ week, exponential DGP, $n_{sim} = 500$). A4 was fitted afterward to the same simulated trials. Two-MCSE band around 0.05: $[0.031, 0.069]$.

Design	N	A0	A1	A2	A3	A4
CO	35	0.150	0.030	0.046	0.052	0.052
CO	70	0.128	0.046	0.054	0.050	0.056
Hybrid	35	0.070	0.042	0.066	0.044	0.048
Hybrid	70	0.060	0.032	0.046	0.054	0.056
OL+BDC	35	0.144	0.048	0.074	0.052	0.052
OL+BDC	70	0.106	0.030	0.046	0.048	0.050

Table S2: Isolation of the three differences between A0 and A1 (simrun job 42, 1,000 replicates per cell, $t_{1/2} = 1$ week, pooled over $N \in \{35, 70\}$, Monte Carlo standard error in parentheses). A0 is the starting analysis. E0n refits it with nlme and no residual correlation. E0nc adds corCAR1 to E0n. E1nc uses the A1 formula with nlme and no residual correlation, that is, without the expectancy term and without the exclusion rule. A1 is E1nc with corCAR1. Power is at $c_{bm} = 0.45$. The calibration ratio κ is the mean reported standard error over the empirical standard deviation, averaged over the two sample sizes. Under CO the expectancy term is absent from A0 because expectancy is constant, so E0n and E1nc are the same model.

Design	Quantity	A0	E0n	E0nc	E1nc	A1
CO	Type I error	0.137 (0.008)	0.137 (0.008)	0.047 (0.005)	0.137 (0.008)	0.046 (0.005)
CO	κ under the null	0.77	0.77	1.00	0.77	1.00
CO	Power	0.821 (0.009)	0.821 (0.009)	0.679 (0.010)	0.821 (0.009)	0.678 (0.010)
Hybrid	Type I error	0.061 (0.005)	0.061 (0.005)	0.034 (0.004)	0.060 (0.005)	0.034 (0.004)
Hybrid	κ under the null	0.95	0.95	1.05	0.96	1.05
Hybrid	Power	0.702 (0.010)	0.702 (0.010)	0.629 (0.011)	0.700 (0.010)	0.631 (0.011)
OL+BDC	Type I error	0.098 (0.007)	0.098 (0.007)	0.034 (0.004)	0.093 (0.007)	0.031 (0.004)
OL+BDC	κ under the null	0.83	0.83	1.07	0.85	1.09
OL+BDC	Power	0.557 (0.011)	0.557 (0.011)	0.399 (0.011)	0.545 (0.011)	0.369 (0.011)

Table S3: Conservatism of the working-covariance interaction test and its cluster-robust correction at four Hybrid null cells ($c_{bm} = 0$, exponential DGP, corrected simulation code, 2,000 replicates each). Each row fits one regressor with the working-covariance standard error (WC) and with the cluster-robust standard error (CR2). The binary row is A1 under WC and A2 under CR2; the CR2 columns of the other rows are A3 and A4. A4 is fitted with CR2 only. κ is the calibration ratio of the main text; $\kappa > 1$ is a conservative test. Two-MCSE band around 0.05: $[0.040, 0.060]$.

$t_{1/2}$	N	Regressor	κ WC	Type I WC	κ CR2	Type I CR2
0.5	35	Binary indicator (A1, A2)	1.05	0.041	0.97	0.049
0.5	35	Exposure-weighted, true $t_{1/2}$ (A3)	1.05	0.035	0.97	0.050
0.5	35	Exposure-weighted, selected $t_{1/2}$ (A4)	–	–	0.96	0.048
0.5	70	Binary indicator (A1, A2)	1.05	0.042	0.98	0.050
0.5	70	Exposure-weighted, true $t_{1/2}$ (A3)	1.06	0.037	0.99	0.050
0.5	70	Exposure-weighted, selected $t_{1/2}$ (A4)	–	–	0.99	0.050
1.0	35	Binary indicator (A1, A2)	1.05	0.040	0.97	0.047
1.0	35	Exposure-weighted, true $t_{1/2}$ (A3)	1.06	0.033	0.98	0.052
1.0	35	Exposure-weighted, selected $t_{1/2}$ (A4)	–	–	0.96	0.053
1.0	70	Binary indicator (A1, A2)	1.05	0.041	0.98	0.049
1.0	70	Exposure-weighted, true $t_{1/2}$ (A3)	1.07	0.035	0.99	0.046
1.0	70	Exposure-weighted, selected $t_{1/2}$ (A4)	–	–	0.99	0.048

Rung 2: the cluster-robust standard error

Table S3 compares the working-covariance and cluster-robust standard errors of each regressor at four Hybrid null cells with 2,000 replicates each.

Rung 4: choosing the half-life

Table S4 gives the half-life that A4 selected. Tables S5 and S6 compare A4 with the oracle A3 in the Hybrid run with independent seeds.

Stress scenarios

Tables S7 and S8 give the single-cell scenarios of the scenario job. Tables S9 and S10 give the scenario grids of the supplementary study.

Heavy missingness

Table S11 gives the dispersion of the estimates in the heavy-missingness stress case.

Table S4: Share of replicates in which A4 selected each candidate half-life (weeks), Hybrid design, pooled over the predictive-only cells of the base grid and the supplementary study at the given true half-life and N (null and alternative cells together).

			Selected half-life			
True $t_{1/2}$	N	n	0.25	0.5	1	2
0.0	35	500	0.97	0.03	0.00	0.00
0.0	70	500	0.99	0.01	0.00	0.00
0.2	35	1000	0.95	0.05	0.00	0.00
0.2	70	991	0.99	0.01	0.00	0.00
0.5	35	13482	0.62	0.27	0.09	0.03
0.5	70	13449	0.63	0.30	0.06	0.01
1.0	35	3497	0.19	0.24	0.29	0.28
1.0	70	3489	0.12	0.27	0.40	0.21
2.0	35	1000	0.14	0.06	0.14	0.65
2.0	70	992	0.05	0.06	0.17	0.73
4.0	35	1000	0.23	0.05	0.09	0.63
4.0	70	990	0.18	0.02	0.06	0.74

Table S5: Hybrid run with independent seeds (family S): null rejection rate ($c_{bm} = 0$) of A3, the oracle at the true half-life, and A4 in four Hybrid null cells, 5,000 replicates each (MCSE about 0.003), with the paired per-replicate difference of A4 against A3 (paired SE 0.0015 to 0.0028) and the calibration ratio.

Cell	Method	Rejection	Paired diff. vs A3	κ
$t_{1/2} = 0.5$, $N = 35$	A3	0.0482	–	0.97
	A4	0.0470	-0.0012	0.97
$t_{1/2} = 0.5$, $N = 70$	A3	0.0514	–	0.99
	A4	0.0510	-0.0004	0.98
$t_{1/2} = 1.0$, $N = 35$	A3	0.0508	–	0.96
	A4	0.0506	-0.0002	0.94
$t_{1/2} = 1.0$, $N = 70$	A3	0.0461	–	1.00
	A4	0.0447	-0.0012	0.99

Table S6: Hybrid run with independent seeds (family S): power and calibration ratio of A3 and A4 at eight Hybrid alternative cells, 1,000 replicates each (MCSE of power at most 0.016).

			Power			κ	
$t_{1/2}$	N	c_{bm}	A3	A4	A4 − A3	A3	A4
0.5	35	0.25	0.291	0.277	-0.014	0.95	0.95
0.5	70	0.25	0.538	0.526	-0.012	0.96	0.94
0.5	35	0.45	0.743	0.737	-0.006	0.98	0.91
0.5	70	0.45	0.966	0.960	-0.006	1.01	0.95
2.0	35	0.25	0.124	0.108	-0.016	0.96	0.95
2.0	70	0.25	0.248	0.239	-0.009	1.02	0.99
2.0	35	0.45	0.384	0.362	-0.022	0.94	0.87
2.0	70	0.45	0.679	0.661	-0.019	1.00	0.91

Table S7: Type I error ($c_{bm} = 0$) at four null scenarios on Hybrid ($N = 70$, $t_{1/2} = 0.5$ week unless stated; OL+BDC row is that design), $n_{sim} = 500$, same trial for every specification each replicate. A4 was fitted to the same trials of the two misspecification cells only (– elsewhere). Two-MCSE band around 0.05: $[0.031, 0.069]$.

Cell	A0	A1	A2	A3	A4
Hybrid	0.060	0.034	0.052	0.046	–
OL+BDC	0.100	0.030	0.051	0.048	–
$t_{1/2} = 2$ wk	0.066	0.034	0.052	0.054	0.056
Weibull $k = 2$ data	0.060	0.032	0.052	0.046	0.052

Table S8: Power at five alternative scenarios (same cells and n_{sim} as the null table). A2 and A3 are the equal-size comparators. A0 is omitted because its test is liberal and its power is not comparable.

Cell	A1	A2	A3	A4
Hybrid, $c_{bm} = 0.45$	0.934	0.962	0.972	–
Hybrid, $c_{bm} = 0.25$	0.510	0.564	0.574	–
OL+BDC, $c_{bm} = 0.45$	0.842	0.865	0.871	–
$t_{1/2} = 2$ wk, $c_{bm} = 0.25$	0.180	0.196	0.257	0.253
Weibull $k = 2$ data, $c_{bm} = 0.25$	0.630	0.653	0.655	0.661

Table S9: Scenario grids: Type I error ($c_{bm} = 0$) on Hybrid at $N \in \{35, 70\}$ under a Weibull decay analyzed as exponential, a true half-life below, at the boundary of and above A4's candidate grid, and two autocorrelations ($t_{1/2} = 0.5$ week and $\rho = 0.7$ unless stated; $n_{sim} = 500$; band $[0.031, 0.069]$).

Scenario	N	A0	A1	A2	A3	A4
Weibull $k = 0.5$ data	35	0.054	0.038	0.040	0.042	0.044
Weibull $k = 0.5$ data	70	0.060	0.026	0.034	0.040	0.044
Weibull $k = 2$ data	35	0.054	0.038	0.040	0.042	0.038
Weibull $k = 2$ data	70	0.062	0.026	0.036	0.042	0.036
$t_{1/2} = 0.2$ wk	35	0.054	0.038	0.040	0.038	0.038
$t_{1/2} = 0.2$ wk	70	0.060	0.026	0.036	0.032	0.034
$t_{1/2} = 2$ wk	35	0.048	0.036	0.042	0.050	0.044
$t_{1/2} = 2$ wk	70	0.056	0.026	0.034	0.046	0.050
$t_{1/2} = 4$ wk	35	0.050	0.036	0.042	0.044	0.042
$t_{1/2} = 4$ wk	70	0.060	0.026	0.036	0.042	0.044
$\rho = 0.5$	35	0.056	0.032	0.044	0.046	0.052
$\rho = 0.5$	70	0.062	0.024	0.030	0.044	0.040
$\rho = 0.9$	35	0.040	0.016	0.038	0.036	0.032
$\rho = 0.9$	70	0.034	0.018	0.044	0.050	0.046

Table S10: Scenario grids: power at one alternative per scenario (same cells and n_{sim} as the null table; $c_{bm} = 0.30$ for the autocorrelation rows because 0.45 is infeasible at $\rho = 0.9$). A0 is omitted because its test is liberal.

Scenario	N	A1	A2	A3	A4
Weibull $k = 0.5$ data, $c_{bm} = 0.45$	35	0.538	0.548	0.586	0.566
Weibull $k = 0.5$ data, $c_{bm} = 0.45$	70	0.878	0.900	0.915	0.907
Weibull $k = 2$ data, $c_{bm} = 0.45$	35	0.830	0.850	0.848	0.850
Weibull $k = 2$ data, $c_{bm} = 0.45$	70	0.994	0.996	0.994	0.994
$t_{1/2} = 0.2$ wk, $c_{bm} = 0.25$	35	0.348	0.340	0.342	0.338
$t_{1/2} = 0.2$ wk, $c_{bm} = 0.25$	70	0.632	0.659	0.659	0.655
$t_{1/2} = 2$ wk, $c_{bm} = 0.25$	35	0.108	0.112	0.146	0.150
$t_{1/2} = 2$ wk, $c_{bm} = 0.25$	70	0.176	0.185	0.233	0.234
$t_{1/2} = 4$ wk, $c_{bm} = 0.25$	35	0.058	0.064	0.080	0.090
$t_{1/2} = 4$ wk, $c_{bm} = 0.25$	70	0.084	0.092	0.134	0.113
$\rho = 0.5$, $c_{bm} = 0.30$	35	0.280	0.284	0.288	0.286
$\rho = 0.5$, $c_{bm} = 0.30$	70	0.542	0.595	0.606	0.585
$\rho = 0.9$, $c_{bm} = 0.30$	35	0.684	0.785	0.808	0.804
$\rho = 0.9$, $c_{bm} = 0.30$	70	0.974	0.994	0.990	0.994

Table S11: Heavy-missingness stress case (per-occasion MCAR 0.3, $N = 35$, 500 replicates): dispersion of the interaction estimates on the calibrated scale and the calibration ratio on the conventional scale (κ, mean SE over SD) and a robust one (median SE over 1.4826 times the median absolute deviation). A4 was not fitted to these cells. From '34-referee-reanalysis.R'.

Cell	Method	SD of estimates	MAD-based SD	Max $\lvert\widehat{\beta}\rvert$	κ	Robust κ	Rejection
null	A0	7.0	3.7	83	0.68	1.03	0.040
	A1	6.1	3.3	82	0.73	1.10	0.026
	A2	6.1	3.3	82	0.61	0.90	0.020
	A3	7.6	4.2	102	0.60	0.88	0.024
$c_{bm} = 0.45$	A0	6.9	3.6	91	0.68	1.05	0.126
	A1	6.1	3.1	89	0.73	1.18	0.103
	A2	6.1	3.1	89	0.58	0.94	0.075
	A3	7.5	3.9	110	0.57	0.91	0.085

Table S12: Patient slope variance (family 5): rejection rate (MCSE), mean interaction estimate and calibration ratio of A3 and A4 under a patient-specific linear trend in the natural-history component (SD 0.3 outcome units per week), Hybrid, $N = 70$, $t_{1/2} = 0.5$ week, 1,000 replicates. A4 was fitted afterward to the same trials. The true value at the alternative is −3.6.

Cell	Method	Rejection	Mean estimate	κ
null ($c_{bm} = 0$)	A3	0.042 (0.006)	-0.02	1.01
	A4	0.042 (0.006)	-0.01	1.00
$c_{bm} = 0.45$	A3	0.970 (0.005)	-3.63	0.98
	A4	0.966 (0.006)	-3.39	0.94

Table S13: Total N for 80 percent power of A4 in the Hybrid design by serial correlation ρ, carryover half-life (weeks) and c_{bm} (columns), under the starting values, from runs made with the companion paper's simulation code (six or seven values of N per cell, a probit curve on $\sqrt{N}$ per cell). The grid of c_{bm} differs by ρ because the largest feasible c_{bm} falls as ρ rises; a blank cell was not simulated.

		c_{bm}				
ρ	$t_{1/2}$	0.20	0.25	0.30	0.35	0.45
0.5	0.0		271		142	83
	0.2		268		140	86
	1.0		434		244	146
0.7	0.0		193		104	61
	0.2		205		108	64
	1.0		384		193	121
0.9	0.0	110	68	50		
	0.2	117	74	53		
	1.0	240	153	107		

Patient slope variance

Table S12 gives A3 and A4 under a patient-specific linear trend in the natural-history component.

Sample size for A4

Table S13 gives the total N for 80 percent power of A4 in the Hybrid design at three values of the serial correlation, from runs made with the companion paper's simulation code under its starting values; the companion paper does not report these runs. Table S14 gives the total N for A2, the binary indicator with CR2, from the same runs as the A4 table of the main text. Table S15 gives the power of A4 at the sample size of the companion paper's closed-form rule. Figure S1 gives the power curves of the main text under the starting values.

Table S14: Total N for 80 percent power of A2, the binary indicator with CR2, at $\alpha = 0.05$, with a 95 percent Monte Carlo interval, by design, carryover half-life (weeks) and c_{bm}, under the parameter set of the main text and under the starting values of the companion paper. Same runs, curve model and conditions as the A4 table of the main text, so the two tables can be compared cell by cell.

		Main text set, c_{bm}			Starting values, c_{bm}		
Design	$t_{1/2}$	0.2	0.3	0.4	0.2	0.3	0.4
CO	0.0	299 (292–307)	133 (130–137)	75 (73–77)	563 (548–579)	250 (244–258)	141 (137–145)
	0.5	299 (291–307)	133 (130–137)	75 (73–77)	570 (556–587)	254 (247–261)	143 (139–147)
	1.0	315 (307–323)	140 (137–144)	79 (77–81)	616 (600–634)	274 (267–282)	154 (150–159)
OL+BDC	0.0	195 (190–200)	87 (85–89)	49 (48–50)	404 (393–416)	180 (175–185)	101 (99–104)
	0.5	307 (300–316)	137 (133–141)	77 (75–79)	637 (619–656)	283 (275–292)	160 (155–164)
	1.0	579 (557–604)	257 (248–269)	145 (140–151)	1231 (1180–1288)	548 (525–573)	308 (295–322)
Hybrid	0.0	151 (146–156)	68 (65–70)	38 (37–39)	310 (301–319)	138 (134–142)	78 (76–80)
	0.5	210 (205–216)	94 (91–96)	53 (52–54)	437 (425–449)	194 (189–200)	110 (107–113)
	1.0	345 (336–355)	154 (150–158)	87 (84–89)	704 (684–725)	313 (304–322)	176 (171–182)

Table S15: Power of A4 at the total N given by the closed-form paired-difference rule of the companion paper for 80 percent power (in parentheses), by design, carryover half-life (weeks) and c_{bm} (columns), under the starting values at $\rho = 0.7$. Runs made with the companion paper's simulation code, 1,000 replicates per cell (Monte Carlo SE at most 0.016). [a]The closed-form N exceeded 400 and the cell was run at $N = 400$, so its power is not at the rule's N.

		c_{bm}		
Design	$t_{1/2}$	0.25	0.35	0.45
CO	0.0	0.842 (400)[a]	0.861 (216)	0.860 (130)
	0.2	0.840 (400)[a]	0.867 (216)	0.869 (130)
	1.0	0.812 (400)[a]	0.862 (230)	0.844 (138)
OL+BDC	0.0	0.936 (394)	0.934 (200)	0.930 (120)
	0.2	0.915 (400)[a]	0.936 (206)	0.932 (124)
	1.0	0.601 (400)[a]	0.866 (400)[a]	0.891 (272)
Hybrid	0.0	0.915 (272)	0.908 (138)	0.894 (82)
	0.2	0.900 (278)	0.899 (140)	0.903 (84)
	1.0	0.809 (400)[a]	0.885 (244)	0.874 (146)

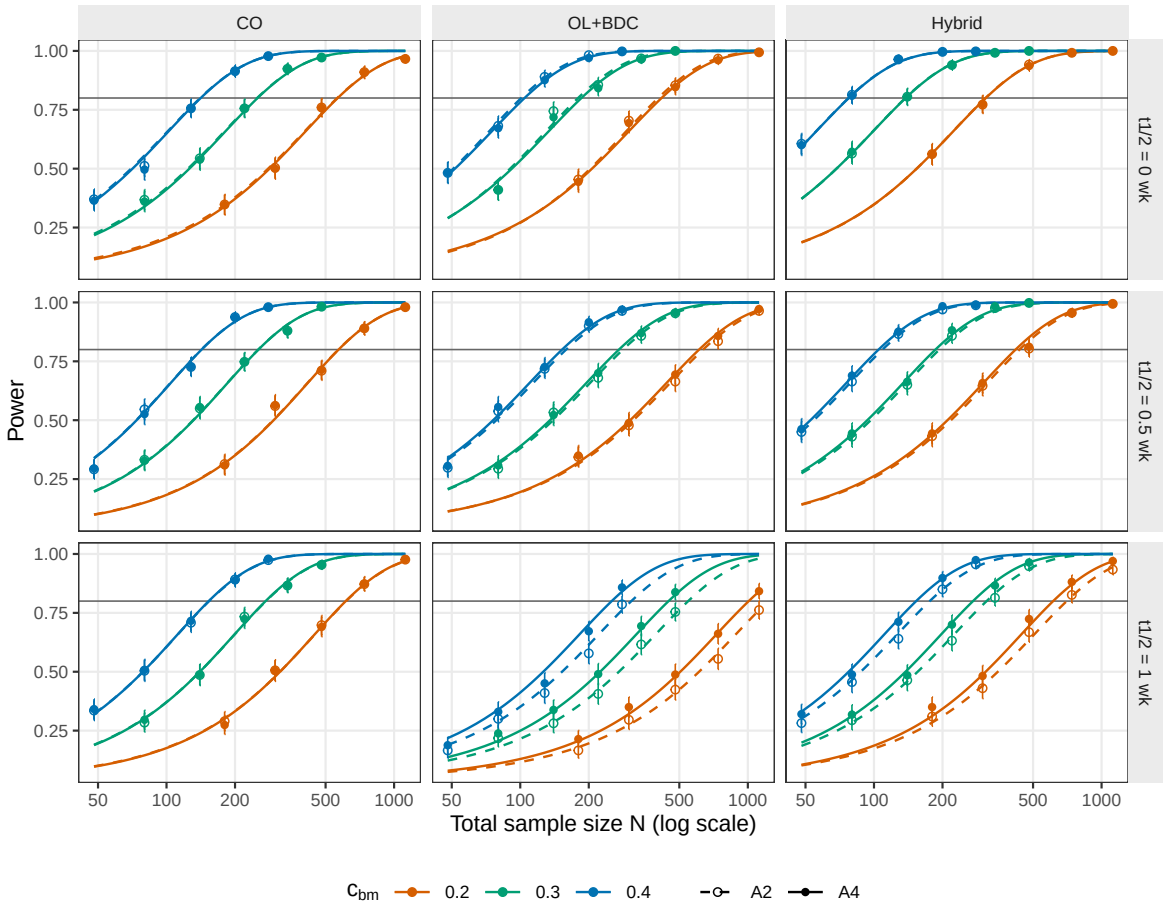


Figure S1: Power of A4 (filled points, solid lines) and A2 (open points, dashed lines) against total N under the starting values of the companion paper, as the sample-size figure of the main text. The horizontal line is 80 percent power.